

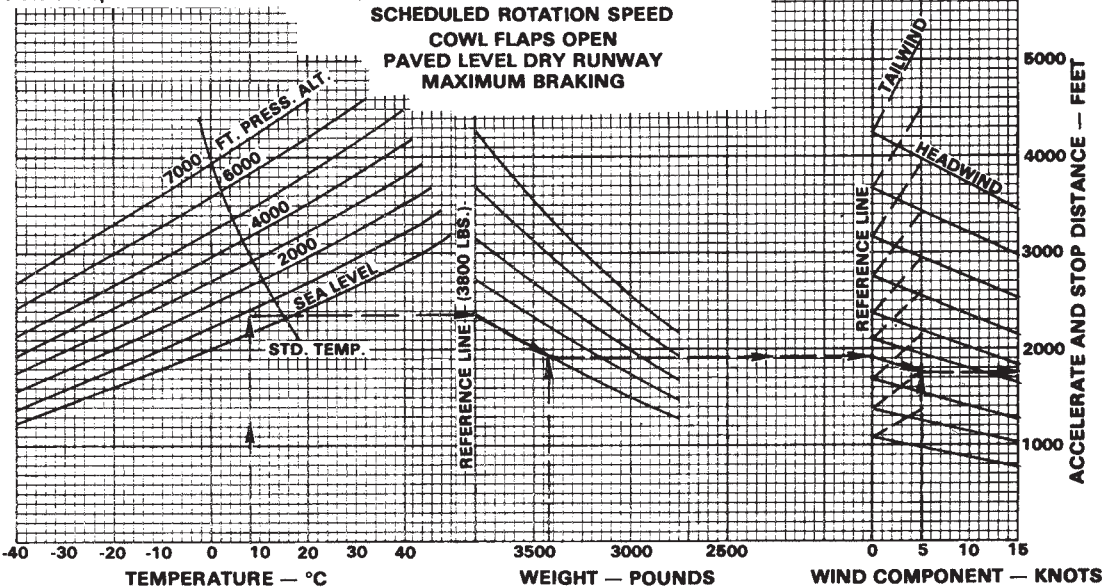
PA-44-180

SHORT FIELD ACCELERATE AND STOP DISTANCE

Example:

Airport press. alt.: 680 ft.
Outside air temp.: 8°C
Weight: 3430 lbs.
Wind component: 5 kts. headwind
Accelerate & stop distance: 1750 ft.

BOTH ENGINES 2700 RPM & FULL THROTTLE
MIXTURE FULL RICH
WING FLAPS 0° - ABORT SPEED AT
SCHEDULED ROTATION SPEED
COWL FLAPS OPEN
PAVED LEVEL DRY RUNWAY
MAXIMUM BRAKING



ACCELERATE AND STOP DISTANCE - SHORT FIELD EFFORT

Figure 5-10

TAKEOFF DISTANCE OVER 50 FT OBSTACLE - SHORT FIELD EFFORT

ASSOCIATED CONDITIONS:

Wing Flaps: 0°
 Cowl Flaps: OPEN
 Power: 2700 RPM & FULL THROTTLE
 BEFORE BRAKE RELEASE
 Runway: PAVED, LEVEL & DRY

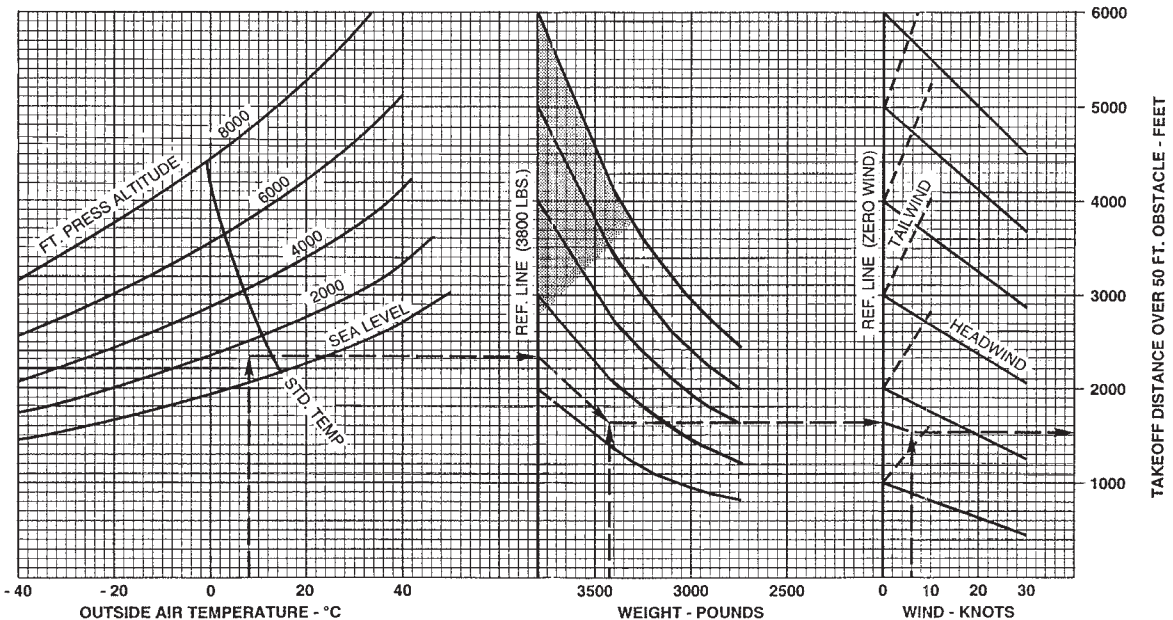
WEIGHT POUNDS	ROTATE SPEED KIAS	OBSTACLE SPEED KIAS
3800	70	82
3400	66	77
3000	62	72
2600	57	67

CAUTION

BEST ONE ENGINE INOPER-
 ATIVE RATE OF CLIMB IS
 LESS THAN 50 FPM IF T.O.
 WT. IS IN THE SHADED AREA

EXAMPLE:

O.A.T.: 8°C
 Airport Pressure Altitude: 1250 FT
 Weight: 3430 LBS
 Wind Component: 6 KT HEADWIND
 Takeoff Distance
 Over 50 FT Obstacle: 1520 FT



TAKEOFF DISTANCE OVER 50 FT. OBSTACLE - SHORT FIELD EFFORT
 Figure 5-13

CLIMB PERFORMANCE - BOTH ENGINES OPERATING - GEAR DOWN

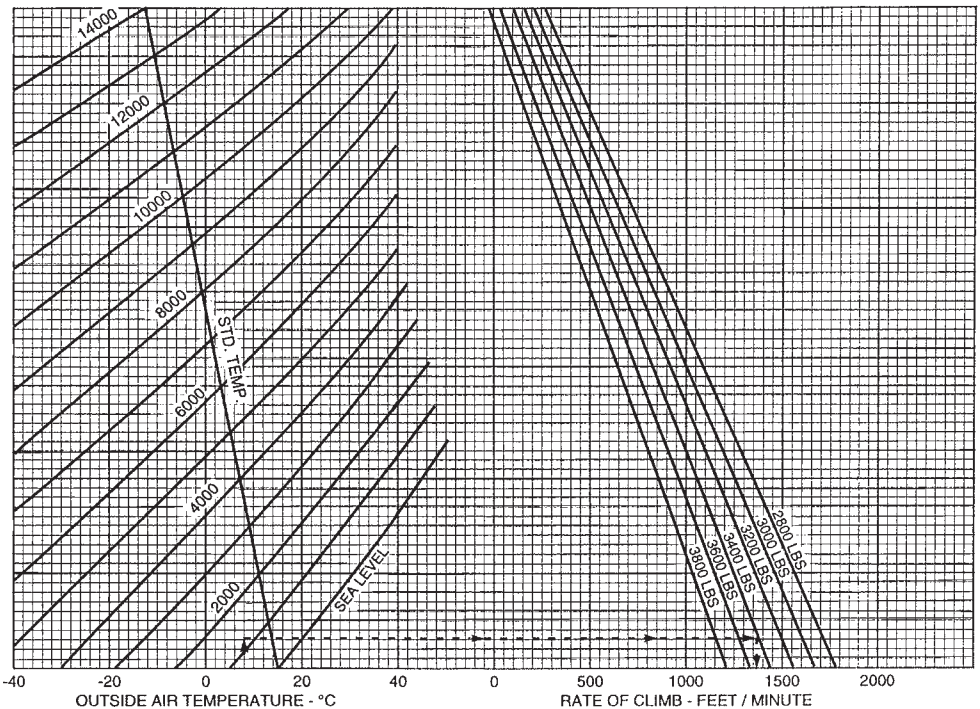
ASSOCIATED CONDITIONS:

Wing Flaps: 0°
 Cowl Flaps: OPEN
 Landing Gear: DOWN
 Mixture: FULL RICH

Engines: BOTH OPERATING
 Power: 2700 RPM
 FULL THROTTLE
 Airspeed: 78 KIAS

EXAMPLE:

Outside Air Temp. 8°C
 Press. Alt.: 1250 FT.
 Weight: 3430 LBS
 Rate of Climb: 1360 F.P.M.



CLIMB PERFORMANCE - BOTH ENGINES OPERATING - GEAR DOWN

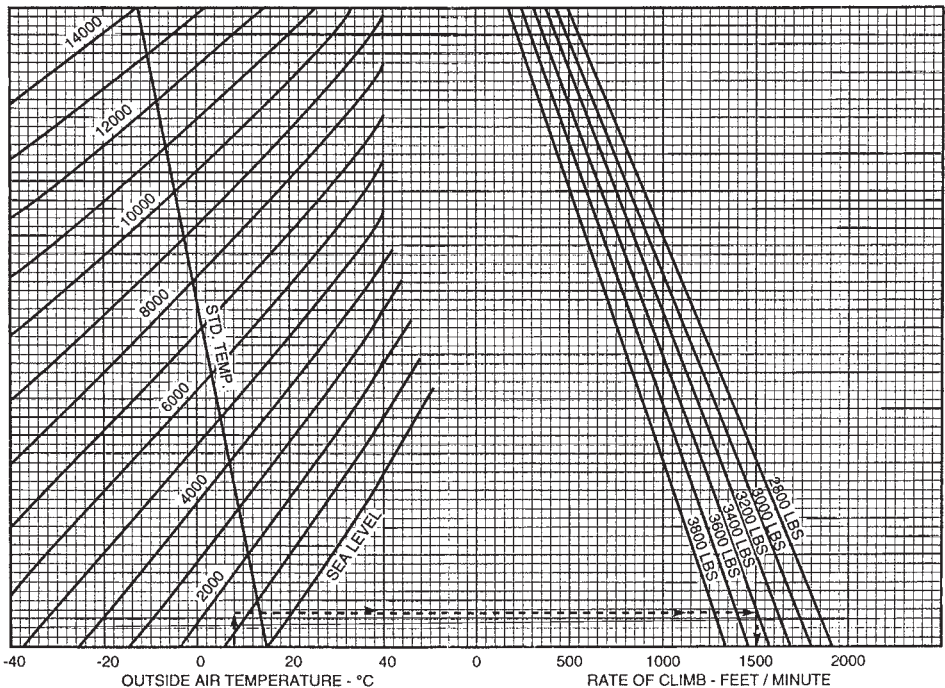
Figure 5-15

CLIMB PERFORMANCE - BOTH ENGINES OPERATING - GEAR UP**ASSOCIATED CONDITIONS:**

Power FULL THROTTLE, 2700 RPM
 Cowl Flaps OPEN
 Mixture FULL RICH
 Airspeed 88 KIAS
 Wing Flaps 0°

Engines: BOTH OPERATING
 Landing Gear: UP

EXAMPLE:
 Press. Alt.: 1250 FT.
 Outside Air Temp. 8°C
 Weight: 3430 LBS
 Rate of Climb: 1505 FT/MIN



CLIMB PERFORMANCE - BOTH ENGINES OPERATING - GEAR UP

Figure 5-17

CLIMB PERFORMANCE - ONE ENGINE OPERATING - GEAR UP

ASSOCIATED CONDITIONS:

Wing Flaps: 0°
Cowl Flaps
(Operating Engine): OPEN
(Inoperative Engine): CLOSED
Landing Gear: UP

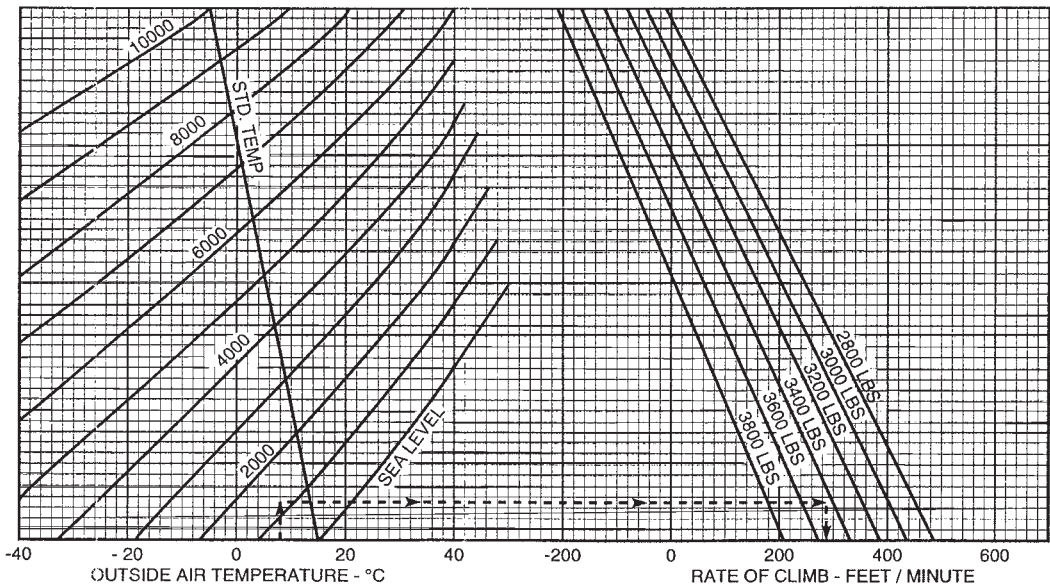
Mixture: FULL RICH
Prop
(Inoperative Engine): FEATHERED
Power: 2700 RPM,
FULL THROTTLE
88 KIAS

NOTE

2° TO 3° BANK TOWARD
OPERATING ENGINE

EXAMPLE:

Outside Air Temp.: 8°C
Press Alt.: 1250 FT.
Weight: 3430
One Engine
Inoperative Climb: 285 F.P.M.



CLIMB PERFORMANCE - ONE ENGINE OPERATING - GEAR UP

Figure 5-19

LANDING DISTANCE OVER 50 FT OBSTACLE — SHORT FIELD EFFORT

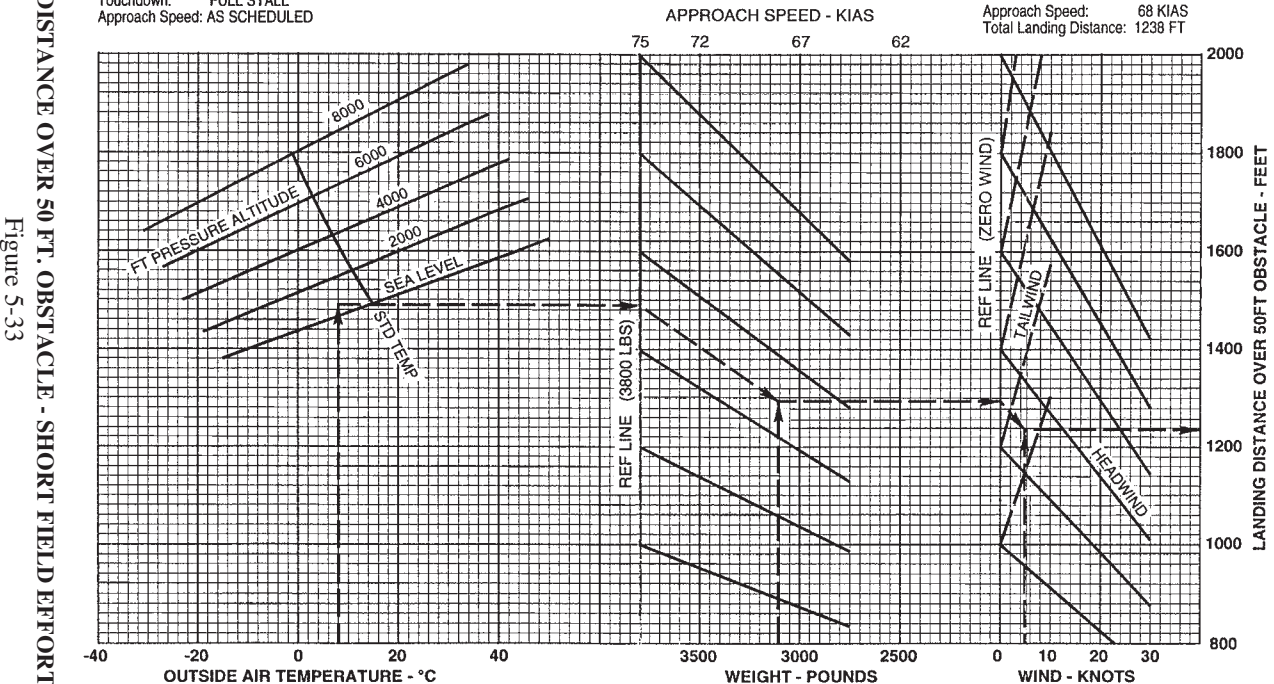
ASSOCIATED CONDITIONS:

Wing Flaps: 40°
 Power: OFF
 Cowl Flaps: AS REQUIRED
 Runway: PAVED LEVEL & DRY
 Touchdown: FULL STALL
 Approach Speed: AS SCHEDULED

EXAMPLE:

O.A.T.: 8°
 Press. Alt.: 680 FT
 Weight: 3107 LBS
 Wind Component: 5 KT HEADWIND

Approach Speed: 68 KIAS
 Total Landing Distance: 1238 FT



LANDING DISTANCE OVER 50 FT. OBSTACLE - SHORT FIELD EFFORT
 Figure 5-33